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CLASS:5 D
LAB 7 SE

STEPS:

Step 1: Installing dependencies:

a. *pip install hypothesis* AND
b. pip install pytest

```
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python -m pytest --version
pytest 9.2.0
```

Step 2: Run the buggy application .py in the repo assigned to you (under Lab7):

Command: python processor.py

Provide the screenshot of the output

```
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7>
```

Step 3: Complete the code in test_processor.py using hypothesis:

Goal: Use Hypothesis to uncover more edge cases automatically for all three

Functions.

```
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7>
```

Step 4: Run the test_processor.py :

Command: pytest test_processor.py **Or**
python -m pytest test_processor.py (if pytest is installed but still showing not recognised) **Provide the screenshot of test cases failing.**

```
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python -m pytest test_daycare_fees.py
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================= short test summary info =========================
FAILED test_daycare_fees.py::test_parse_enrolment_no_crash - ExceptionGroup: Hypothes
ound 3 distinct failures. (3 sub-exceptions)
FAILED test_daycare_fees.py::test_hours_average_no_crash - ExceptionGroup: Hypothesi
nd 2 distinct failures. (2 sub-exceptions)
FAILED test_daycare_fees.py::test_payable_fee_no_crash - ExceptionGroup: Hypothesis
 3 distinct failures. (3 sub-exceptions)
====================== 3 failed, 1 warning in 4.20s =======================
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7>
```

Step 5: Fix the buggy application python script:

Goal: Harden the functions against all unexpected or invalid inputs, especially those discovered through fuzz testing.

Run the fixed processor.py – Command : python processor.py

Provide the screenshot of the output.

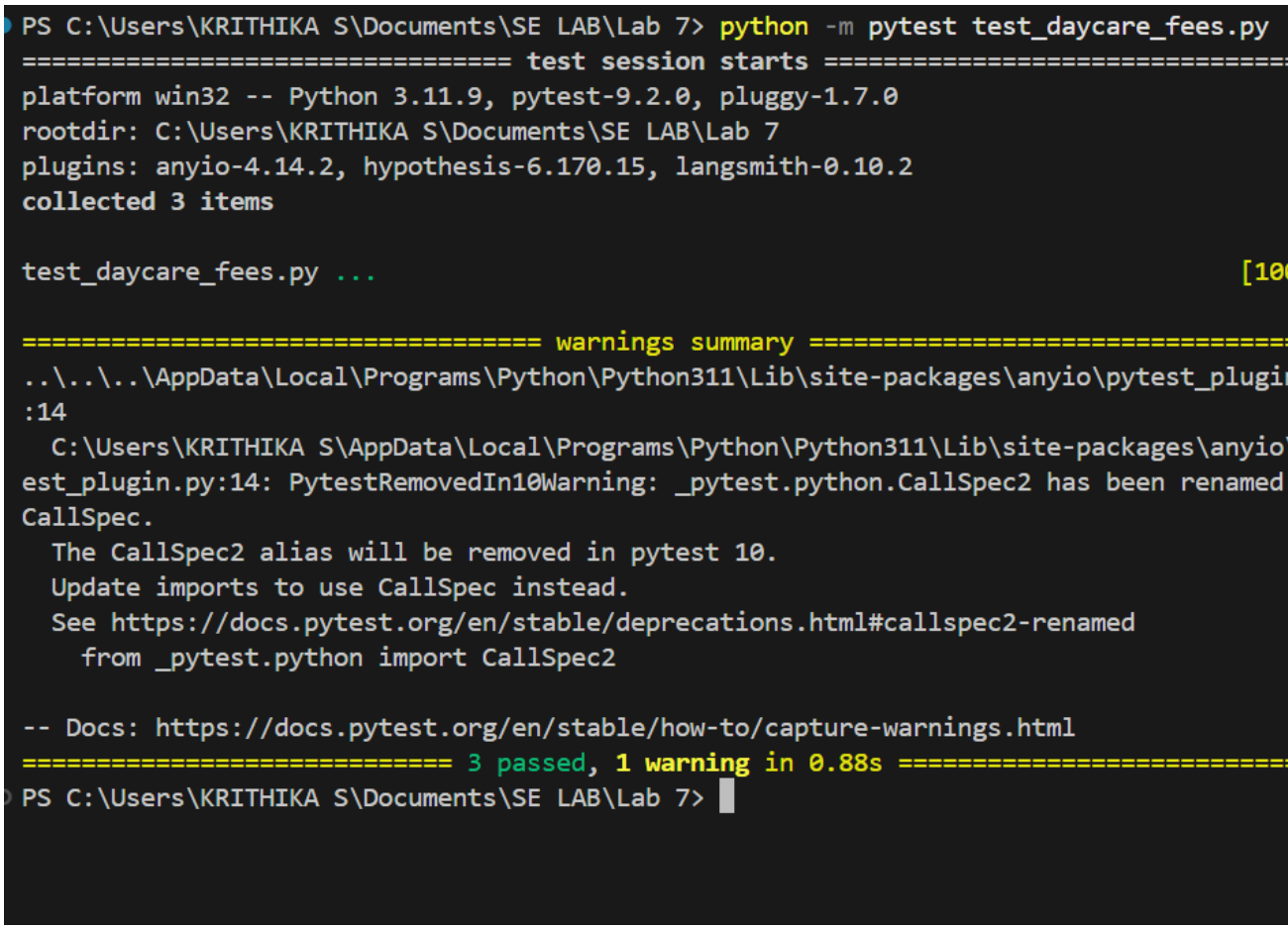
```
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 20,25,30
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7>
```

Step 6: Re-Run the test_processor.py wrt to fixed processor and take the ss of the output:

Command: pytest test_processor.py **Or**
python -m pytest test_processor.py (if pytest is installed but still showing not recognised) **Provide the screenshot of the output.**


PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7> python -m pytest test_daycare_fees.py
test session starts
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.15, langsmith-0.10.2
collected 3 items
test_daycare_fees.py ...
warnings summary
..\..\..\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio\pytest_plugi
:14
C:\Users\KRITHIKA S\AppData\Local\Programs\Python\Python311\Lib\site-packages\anyio
est_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed
CallSpec.
The CallSpec2 alias will be removed in pytest 10.
Update imports to use CallSpec instead.
See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
from _pytest.python import CallSpec2
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
3 passed, 1 warning in 0.88s
PS C:\Users\KRITHIKA S\Documents\SE LAB\Lab 7>